

eKey – JVP Process Interface

Setup and configuration help

Overview

This JUNG Visu Pro (JVP) process interface connects eKey fingerprint access systems to JVP. It supports two independent input channels that can be used at the same time:

- **eKey net / Home / Multi + Converter UDP/LAN** (local, no cloud)
- **ekey bionyx** (cloud, “function webhook”)

Both channels can be active in parallel or used individually. The interface lets you create any number of *Function* instances (one per door/action) that set their **TRIGGERED** output for a configurable pulse duration whenever a matching event is received.

Installation

- Copy the whole **eKey** folder to C:\Users\Public\Documents\JUNG Visu Pro\App\EsfSdvEditor\Interfaces\eKey.
- Restart the JVP device editor (new templates are only read in at startup).
- Choose the **eKey** template under *File / New process interface*.
- Create one **Function** instance per door/action.

Channel 1: eKey net / Home / Multi + Converter UDP/LAN

The eKey Converter UDP/LAN sends a UDP datagram to a configured target IP and port whenever a finger is scanned.

Configure on the eKey Converter (“ekey ConfigConverterUDP” tool)

- Protocol: **home** or **multi** (the binary **rare** format is not supported).
- Separator: freely selectable, default **?**.
- Target IP address: the IP of the machine running JVP.
- Target port: must match **LOCALPORT** of the UDPPORT resource in InterfaceDescription.xml (default **41000**).
- Firewall: the chosen port must accept inbound UDP.

Configure in JVP (CONFIGURATION folder)

- **UDP_PROTOCOL**: home or multi, must match the converter setting.
- **UDP_SEPARATOR**: must match the separator configured on the converter.

Create one Function instance per door/person

- **MATCH_USERID**: the UserID that shall trigger this function.
- **MATCH_FINGERID** (optional): only trigger for this specific finger.
- **MATCH_ACTIONCODE** (optional): only trigger for this action code.

Important note on field order/action codes: The mapping Packet type, UserID, Finger ID, Scanner serial, Action code, Relay ID (home format) resp. Packet type, UserID, User name, User status, Finger ID, Key ID,

Scanner serial, Scanner name, Action code, Input ID (multi format) is derived from the eKey Converter manual and one example telegram printed there. The exact meaning of action codes 1–8 is not fully documented there and may differ by converter firmware. Verify against a real finger scan before going live: STATUS.LAST_RAW and STATUS.LAST_ACTION_CODE show the received raw data.

Channel 2: ekey bionyx (cloud webhook)

ekey bionyx calls a freely configurable HTTP request (“function webhook”), defined in the ekey app, whenever a configured biometric function fires. This interface provides an embedded HTTP server to receive it.

Set up in the ekey bionyx app (Third-Party API / function webhook)

- Method: **POST**
- URL: `http://<JVP-host-IP>:<port>/ekeybionyx` (port = LOCALPORT of the HTTPSERVER resource, default **8090**).
- Header: Authorization: Bearer <token> – must match CONFIGURATION.BIONYX_TOKEN in JVP.
- Body (JSON), a flat object with the key “function”, see example below.

```
{"function": "<name>", "user": "<user>", "finger": "<finger>",  
  "terminal": "<terminal>"}
```

If the JVP machine is only reachable from the local network (recommended), the local IP is enough. If bionyx (cloud) needs to reach the webhook over the internet, port forwarding **and** TLS are mandatory (set USESSL="1" and CERTFILE in the XML).

Configure in JVP

- **CONFIGURATION.BIONYX_TOKEN**: same value as in the Authorization header of the ekey app. Leave empty to disable the check (testing only).

Create one Function instance per function

- **MATCH_BIONYX_FUNCTION**: the value the ekey app sends in the “function” field of the webhook body.

Shared datapoints

- **STATUS.LAST_SOURCE**: UDP or BIONYX – which channel delivered the last event.
- **STATUS.LAST_***: details of the last event (user, finger, serial, action code, raw data, timestamp).
- **Function.TRIGGERED**: pulses to 1 for Function.PULSE_MS milliseconds (about 1 s resolution) when a matching event arrives.

Security notes

- Plain HTTP for the bionyx webhook is only acceptable on a trusted local network. If reachable from the internet, always enable USESSL="1" with a valid certificate **and** a bearer token.
- The bearer token is checked by the script itself (not via the XML attributes USEBEARERTOKEN/BEARERTOKEN), so it can be managed via the CONFIGURATION.BIONYX_TOKEN datapoint without editing the XML.

Known limitations

- The binary **rare** protocol of the eKey Converter is not supported (only home/multi).
- The JSON parser for bionyx webhooks is intentionally simple (flat object, no nested structures, no escaped quotes in values).
- The pulse duration (PULSE_MS) has a resolution of about 1 second.
- The exact meaning of the UDP action codes must be verified against your own system.